Python Zero to Projects

Chapter 2

Practice Set

Variables & Data Types

Instructions: Each question below is a coding task. Write and run each answer as its own separate Python file. Do not attempt to write answers here. There are no hints, solutions, or expected outputs in this document - figure it out yourself by writing, running, and testing your code.

Question 1 - Personal Info Card

Write a program that stores the following information about yourself in separate variables: your first name, your last name, your age, your height in metres (as a decimal number), and whether you are a student (True or False). Then use print() to display each value on its own line, along with a label and its data type using type(). Make sure your variable names follow PEP 8 snake_case convention.

Question 2 - Dynamic Typing Demo

Write a program that demonstrates Python's dynamic typing. Create a single variable called "data" and assign it each of the following values one at a time, printing the value and its type after each assignment: an integer, a float, a string, and a boolean. The program should print 4 lines total, each showing the current value and its type.

Question 3 - User Greeting

Write a program that asks the user to enter their first name and their favourite number (as an integer). Store both inputs in appropriately named variables. Then print a personalised greeting that includes their name, their favourite number, and that favourite number multiplied by 2. Use type() to confirm that the favourite number was successfully converted to an integer.

Question 4 - Type Casting Chain

Write a program that starts with the string "7" stored in a variable called source. Using that single variable as the starting point, create three new variables by casting source to int, float, and back to str. Print all four variables (source plus the three new ones) along with their types. Then try to cast the string "abc" to int inside the same program - observe and comment in your code what error this produces when you run it. (You may comment out the line after observing the error so the rest of the program runs.)

Question 5 - Simple Calculator from Input

Write a program that asks the user to enter two decimal numbers. Cast both inputs to float immediately after reading them. Then print the result of adding the two numbers together, making sure the output clearly shows the user which two numbers were added and what the result is. Demonstrate that without casting, the + operator on two string inputs would concatenate instead of adding (show this in a comment in your code).

Question 6 - Naming Rules Audit

Write a program that creates exactly six variables: two with valid PEP 8 snake_case names, two with valid names that are technically legal but do NOT follow PEP 8 convention (e.g., camelCase or ALL_CAPS used for a non-constant), and two constants written in the correct UPPERCASE convention. Assign any sensible values to each. Print all six variables with a label identifying which category each one belongs to.

Question 7 - Multiple Assignment and Swap

Write a program that does all of the following in order: assign three variables (x, y, z) to the values 100, 200, 300 using a single line of multiple assignment; print x, y, z; then swap the values of x and z using Python's single-line swap syntax; print x, y, z again to confirm the swap; then use the same-value multiple assignment to set all three variables to 0 and print them a final time.

Question 8 - Input Type Trap

Write a program that asks the user to enter two whole numbers. In the first version, do NOT cast the input - store both as strings and print their "sum" using +. In the second version, cast both inputs to int and print their real arithmetic sum. Both versions should be in the same file, clearly separated by comments explaining what is happening in each case and why the outputs differ.

Question 9 - Data Type Detective

Write a program that creates 8 variables with a variety of values including: at least one positive int, one negative int, one float with a decimal, one float in scientific notation (e.g. 1.5e3), one single-character string, one multi-word string, one True boolean, and one False boolean. For each variable, print one line showing the variable's value and its type using type(). The program should produce exactly 8 output lines.

Question 10 - Mini User Profile Builder

Write a program that collects the following information from the user via input(): full name (str), age (int), height in cm (float), and years of Python experience (int). After collecting all four inputs with appropriate type casting, print a neatly formatted profile card. The card must include all four values, must NOT use any f-strings (use string concatenation with str() instead), and must show the data type of each field using type(). Finally, compute and print the user's height in metres (divide cm by 100) as a float.
